MIL-B-52209A(ME)
31 July 1972
SUPERSEDING
MIL-B-52209(CE)
7 February 1962

MILITARY SPECIFICATION

BOLT ASSEMBLIES, HOOK, TIE: WITH HEXAGON NUT AND TRIANGULAR WASHER

1. SCOPE

1.1 Scope. This specification covers tie hook bolt assemblies to be used on military railway bridges.

1.2 Classification. The tie hook bolt assemblies shall be of the following lengths, as specified (see 6.2):

11-1/2 inches
13-1/2 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castellated, Clinch, Knurled and Welding.
PPP-H-1581	- Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

FSC 5420

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A325 - High Strength Bolts for Structural Steel Joints Including Suitable Nuts and Plain Hardened Washers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Description. The tie hook bolt assemblies shall be as shown on figure 1 and as specified herein.

3.2 First article (preproduction model). The supplier shall furnish one tie hook bolt assembly of each length, as applicable, for examination within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce tie hook bolt assemblies that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Bolts. Bolts shall conform to ASTM A325

3.5 Triangular washers. The material and hardness of the triangular washers shall conform to ASTM A325.

3.6 Nuts. Nuts shall conform to FF-N-836, type II, style 4, steel.

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3.7 Screw threads. Screw threads shall conform to Handbook H28,UNC-2A for bolt threads and UNC-2B for nut threads.

3.8 Identification marking. Each tie hook bolt assembly shall be metal-die stamped as shown on the figure. Die stamping shall be legible.

3.9 Workmanship. Steel may be sheared, sawed, or flame-cut. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All burrs and projections shall be removed from holes and edges of components. Components shall be free of scale, fins, porosity, twists, bends, and cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. The tie hook bolt assemblies which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction tie hook bolt assembly(s) shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. Tie hook bolt assemblies shall be examined as specified herein for the following defects:

Major

101. Nonconformance to the dimensions shown on figure 1.
102. Material not as specified.
103. Components missing or not as specified.
104. Components containing twists, bends, burrs, scale, cracks, or other imperfections.

Minor

201. Identification marking incorrect, missing, or illegible.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing and marking shall be examined and tested in accordance with PPP-H-1581.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The tie hook bolt assemblies shall be preserved (using type P-1 preservative), packaged, packed, and marked in accordance with PPP-H-1581. Fiberboard boxes when used for level B packing shall be class weather resistant. The level of preservation and packaging shall be level A or C, and the level of packing shall be level A,B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. The tie hook bolt assemblies are intended for use in securing the rails on fixed railway bridges.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Length of bolt assembly required (see 1.2).
(c) Time frame required for submission of preproduction model (s) (see 3.2).
(d) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production tie hook bolt assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish tie hook bolt assemblies conforming to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A173

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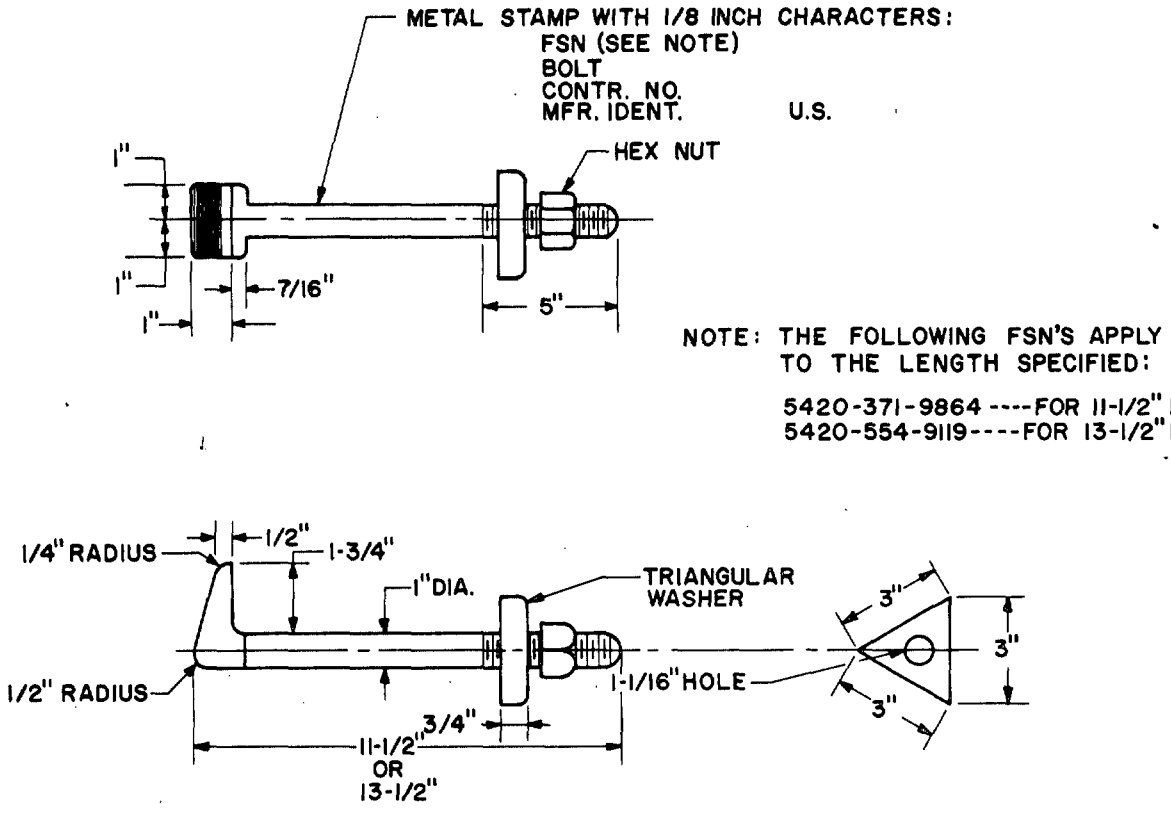


FIGURE 1. TIE HOOK BOLT ASSEMBLY

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-52209A(ME) Bolt Assemblies, Hook, Tie: with Hexagon Nut and Triangular Washer

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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